



Universitat
de les Illes Balears

TRABAJO DE FIN DE MÁSTER

HYDROLOGICAL MODELLING OF THE SANT MIQUEL BASIN: A COMPARATIVE STUDY OF PHYSICS-BASED AND MACHINE LEARNING APPROACHES

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**Máster Universitario en Sistemas Inteligentes (MUSI)
Especialización en Data Science.**

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Palabras clave:

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Hydrological Modelling of the Sant Miquel Basin: A Comparative Study of Physics-Based and Machine Learning Approaches

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Abstract—This manuscript reports the completed data-engineering and classical forecasting stage of a study of water-level prediction in the Sant Miquel basin, Mallorca, Spain. The target is water level at STM08 (Sa Marjal), a coastal wetland at the basin outlet, on a uniform 10-minute grid. A reproducible pipeline combines historical Excel observations, AEMET rainfall records, and a production-database extension. The latter required conversion of water levels from centimetres to metres, quality-aware filtering, and a station-wise correction for anomalous diurnal sensor drift. The resulting dataset contains 618 769 timestamps, 614 201 supervised samples, and 255 predictor variables. Six baseline families are evaluated at 1-hour, 6-hour, and 24-hour horizons: persistence, ridge regression, random forest, XGBoost, ARIMA, and ARIMAX. Persistence is the strongest aggregate baseline, with test NSE values of 0.988, 0.913, and 0.563 at the three horizons. Ridge is the strongest non-persistence model in aggregate, while ARIMA and leakage-free ARIMAX provide more balanced long-horizon errors. The HEC-HMS project has been initialized but not yet calibrated, and sequential neural models have not yet been evaluated; therefore this version makes no physical-versus-deep learning performance claim. It establishes the validated data basis and benchmark results required for those subsequent phases.

I. INTRODUCTION

Flash floods rank among the most destructive natural hazards in Mediterranean Europe, where intense rainfall combines with short concentration times and steep catchments to produce rapid rises and recessions [7]. The Balearic Islands are particularly exposed to high-intensity convective storms and torrential runoff [5]. Reliable short-term water-level prediction is consequently relevant both to civil protection and to the management of wetlands receiving intermittent catchment flows.

The Sant Miquel basin drains toward the Sa Marjal wetland, monitored at station STM08. The basin is instrumented by a dense network of hydrological and meteorological stations, but the data combine different sampling frequencies, long sensor outages, quality flags, and changes in the data acquisition system. These issues are not secondary implementation details: the target is near zero for much of the record, while the operationally important flood response occupies a small fraction of the samples.

Two modelling paradigms motivate the overall thesis. Physics-based models represent rainfall-runoff and routing

processes explicitly and can exploit catchment properties [1]. Data-driven models learn a mapping from observed inputs to future levels; tree ensembles and regularised regression provide transparent benchmarks, while sequential neural architectures are intended for a later phase [2, 6]. The complete thesis will compare these paradigms. The present version documents the finished preprocessing and baseline stage and does not present results for the uncalibrated HEC-HMS setup or for LSTM, GRU, or TFT models.

The current contribution is fourfold: (i) a reproducible 10-minute dataset construction pipeline; (ii) an explicit treatment of the centimetre-based production-database extension and suspicious records; (iii) an event catalogue based on a reproducible peak-over-threshold rule; and (iv) a split-aware benchmark of six data-driven baseline models. These results define the reference performance that future physical and sequential models must exceed, particularly during flood events and at long lead times.

II. STUDY AREA

The Sant Miquel basin is located in northern Mallorca, Spain (approximately 39.80°N, 2.95°E), within the Serra de Tramuntana mountain range. The catchment is underlain primarily by Mesozoic karstic limestone, so part of the drainage network is subsurface. Topography ranges from near sea level at the Sa Marjal wetland to more than 600 m in the headwaters, with steep channel gradients.

The climate is Mediterranean, with most precipitation occurring between October and March and hot, dry summers. Annual rainfall varies spatially from approximately 800 mm in the highlands to 400–500 mm near the coast. The hydrological regime is intermittent: the main channel and tributaries respond rapidly to rainfall and have limited baseflow because of karstic drainage.

Sa Marjal is the terminal wetland of the catchment and the location of STM08, the target station. Near-real-time observations are available through the RiscBal hydrological-risk observatory of the Universitat de les Illes Balears (<https://riscbal.uib.es>), which is the intended operational context of the forecasting system.

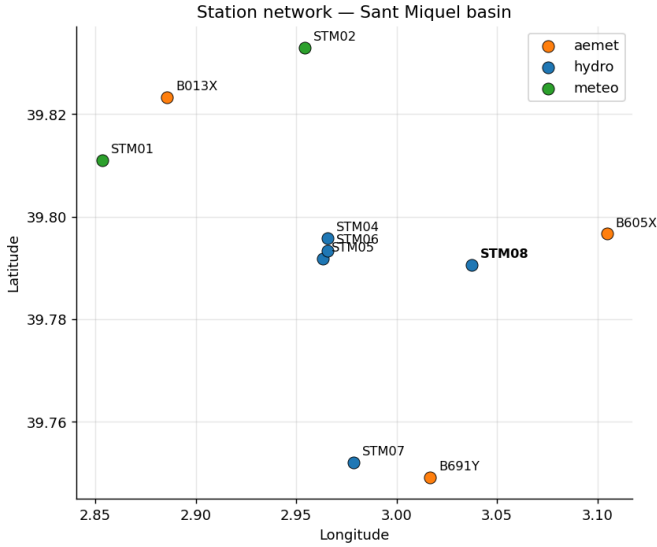


Figure 1. Monitoring network used in the current dataset. STM08 is the target station at the basin outlet.

III. DATA AND MONITORING NETWORK

A. Monitoring stations

The network contains eight STM stations operated by RiscBal and three AEMET rain gauges. STM01 and STM02 are meteorological stations. STM03–STM08 are hydrological stations measuring water level; discharge is derived from level through station-specific rating curves. Water temperature is captured at the hydrological stations when available, but is not used as a predictor because of long, station-wide outages and a substantial availability shift between the chronological splits.

Figure 1 shows the spatial arrangement of the network. STM08 is separated from the upstream hydrological cluster and receives flow through the main channel and a direct tributary.

B. Sources and common time window

Historical STM observations are supplied in Excel files, while the AEMET series combine UIB-Estrany historical records with production-database exports. The hydro-station extension is supplied as station-specific production-database CSV files. The database field named `WaterLevel` is expressed in centimetres; the extension loader converts it to metres before it is written to the common `HEIGHT_m` schema. The conversion is verified against overlapping observations at identical timestamps.

The common modelling window is 2014-09-26 to 2026-07-02. The final date is set by the AEMET series, although some STM and database files continue to 2026-08-05. All timestamps are treated as UTC. The resulting uniform grid has 618 769 timestamps, corresponding to approximately 11.8 years at 10-minute resolution.

C. Data quality and preprocessing

1) *Timestamp alignment and resampling*: The STM loggers contain sub-minute timestamp offsets that drift over time.

Before resampling, timestamps are rounded to the nearest minute and duplicate timestamps are deduplicated. This step is essential: using exact-minute frequency alignment previously discarded true observations and replaced flood peaks with interpolated values. At STM08, correcting this issue restored the historical maximum from 1.31 m to 2.83 m and the corresponding discharge from 19.81 to 87.84 m³ s⁻¹.

Instantaneous variables (level and temperature) are handled as follows: 5-minute records are averaged into 10-minute windows, 15-minute records are linearly interpolated with a three-hour gap cap, and already 10-minute records are directly aligned. Precipitation is treated as an accumulation: the source increments are converted into a cumulative curve, interpolated to the target grid, and differentiated back. This conserves rainfall mass. Hourly AEMET rainfall is disaggregated using contemporaneous STM01/STM02 10-minute templates, with cumulative interpolation as a conservative fallback.

2) *Quality flags and variable selection*: The quality vocabulary is: 0 (good), 1 (suspicious), 2 (wrong), and empty (not supplied for historical Excel records). Production-database hydro records with quality 2 are discarded at ingestion. Quality-1 records are retained, but are explicitly audited. The extension contains 51 336 quality-1 STM03 records and 6 888 quality-1 STM05 records. Their transitions to neighbouring quality-0 data have maximum level changes of 3 mm and 35 mm, respectively, so no discontinuity-based exclusion is applied.

Water temperature remains in the clean files, grid, and measurements table, but is excluded from the training table. STM08 water temperature is absent for 2016–2023 and the validation split has near-complete outages for some water-temperature stations. Similarly, STM02 air temperature is excluded because it is weakly seasonal and absent throughout validation. These decisions avoid training-to-test availability shifts.

Short gaps of up to 24 hours are interpolated. Longer gaps remain NaN and are paired with a missingness indicator. The final imputation pass fills 21 156 cells. Discharge is not imputed because a missing discharge value can mean that the level is outside the rating-curve domain rather than that the sensor failed.

3) *Extension harmonization*: After centimetre-to-metre conversion and quality filtering, the extension is processed using absolute bounds $[-0.5, 5.0]$ m, a maximum change of 0.5 m per 10-minute step, and a minimum valid block of 24 hours with at least 70% valid values. Only 32 non-null values are removed by these safeguards after the unit correction.

The extension also exhibits a station-dependent daily oscillation. The correction procedure is applied uniformly to all six hydro stations. For quiet rows ($H < 0.10$ m), an hour-of-day climatology in the DB-only segment is compared with a historical climatology restricted to the same calendar months. The centred difference is subtracted from the DB-only series only when its amplitude is at least 5 mm and the DB-only range is at least five times the historical range. This preserves a station's established diurnal cycle while removing an anomalous thermal-drift component. Corrected source rows are marked as `DATA_TYPE = corrected` in the clean CSVs.

Table I
MONITORING STATION INVENTORY AND POSITION RELATIVE TO STM08.

ID	Type	Variables	Location	Alt. (m)	Frequency	Position
STM01	Meteorological	PRECIP, TEMP	Coll des Telègraf	1105	15→10 min	15.9 km W
STM02	Meteorological	PRECIP, TEMP	Míner Gran	559	15→10 min	8.5 km WNW
STM03	Hydrological	HEIGHT, WATER TEMP, Q	Es Fangar	73	15→10→5 min	5.1 km WNW
STM04	Hydrological	HEIGHT, WATER TEMP, Q	Gabellí	50	15→10→5 min	6.2 km W
STM05	Hydrological	HEIGHT, WATER TEMP, Q	Monnàber	51	15→10→5 min	6.4 km W
STM06	Hydrological	HEIGHT, WATER TEMP, Q	Sant Miquel	47	15→10→5 min	6.2 km W
STM07	Hydrological	HEIGHT, WATER TEMP, Q	Búger	55	15→10→5 min	6.6 km SW
STM08	Hydrological	HEIGHT, WATER TEMP, Q	Sa Marjal (target)	6	15→10→5 min	basin outlet
B013X	AEMET rain gauge	PRECIP	Lluc	490	60 min	13.5 km WNW
B605X	AEMET rain gauge	PRECIP	Muro-S'Albufera	2	60 min	5.8 km E
B691Y	AEMET rain gauge	PRECIP	Sa Pobla-Sa Canova	40	60 min	4.9 km SSW

Table II
HYDROLOGICAL DB EXTENSION AFTER UNIT AND QUALITY CORRECTIONS. THE FINAL COLUMN IS THE NUMBER SURVIVING ALL HARMONIZATION SAFEGUARDS.

Station	Ext. rows	Non-null	After	Diurnal correction
STM03	102 604	48 531	48 528	no
STM04	82 759	82 759	82 758	yes
STM05	72 424	57 643	57 616	yes
STM06	83 737	83 737	83 736	no
STM07	42 292	42 292	42 292	yes
STM08	46 066	46 066	46 066	yes
Total	429 882	361 028	360 996	

The diurnal criterion is not triggered for STM03 or STM06. STM03 has a substantial cycle that is similar to its historical behaviour and is therefore not removed. STM04 and STM05 meet the criterion under the quiet-row, season-matched calculation. The source-level correction is retained for traceability; the grid's per-variable provenance labels identify re-gridded, derived, and imputed values, while the original correction remains available in the clean CSVs.

IV. DATASET CONSTRUCTION

A. Wide grid and measurements table

The base grid contains 61 columns: 19 measured variables (level, water temperature, air temperature, and precipitation), six discharge series derived from rating curves, quality flags, and provenance columns. The imputed grid contains 80 columns after adding missingness indicators. The long-format advisor table, `measurements.parquet`, contains 15 469 225 rows and six columns: timestamp, station, variable, value, quality, and data type. Its data-type counts are 9 883 004 observed, 5 565 065 derived, and 21 156 imputed records.

The confirmed drainage topology is:

$$\begin{aligned} \text{STM03} &\longrightarrow \text{STM04} \longrightarrow \text{STM06} \longrightarrow \text{STM08}, \\ \text{STM05} &\longrightarrow \text{STM06}, \quad \text{STM07} \longrightarrow \text{STM08}. \end{aligned}$$

Therefore, the engineered total inflow to STM08 is $Q_{\text{IN}} = Q_{06} + Q_{07}$, rather than the earlier sum of all five upstream stations. Two lateral-inflow features are also retained: $\Delta Q_{03,04} = Q_{04} - Q_{03}$ and $\Delta Q_{05,06} = Q_{06} - Q_{04} - Q_{05}$.

Table III
CHRONOLOGICAL SUPERVISED-LEARNING SPLIT.

Split	Period	Rows	Share
Train	Sep. 2014–Dec. 2020	325 049	52.9%
Validation	Jan. 2021–Jun. 2022	78 624	12.8%
Test	Jul. 2022–Jul. 2026	210 528	34.3%

B. Predictors, targets, and splits

The training table contains 614 201 rows and 289 columns, including 255 predictors, three future targets, the chronological split, and event labels. Predictor lags are generated at 1, 2, 3, 6, 12, and 24 hours for the six hydrological levels, STM01 air temperature, the five precipitation stations, and upstream discharge. Cumulative precipitation is computed over 3, 6, 12, 24, and 48 hours. The table also contains temporal sine/cosine encodings, individual upstream-discharge lags, lateral inflows, and total inflow to STM08. Water temperature and STM02 air temperature are not included as predictors for the availability reasons described above.

The target columns are the STM08 level at 1, 6, and 24 hours ahead, equivalent to 6, 36, and 144 ten-minute steps. The split is strictly chronological:

Figure 3 illustrates the exploratory hydrological relationship that motivated the inclusion of upstream levels and flows. STM04 and STM06 are the strongest level predictors of STM08 in the historical cross-correlation analysis, with approximate lags of 30 and 60 minutes respectively.

V. FLOOD EVENT DEFINITION

Flood events are detected on the STM08 discharge series using a peak-over-threshold procedure with an independence criterion [3, 4]. The event core begins when $Q > 0.2 \text{ m}^3 \text{ s}^{-1}$. Consecutive cores with gaps of up to three hours are grouped, and groups separated by less than six hours are merged. Boundaries are expanded using $Q > 0.05 \text{ m}^3 \text{ s}^{-1}$, with up to 12 hours of lookback and 24 hours of lookahead. Cores are capped at 72 hours.

As a final global validity criterion, an event must have $H_{\text{peak}} \geq 0.10 \text{ m}$. This criterion is applied to all years and data sources, not only to the database extension. It removes minor pulses and the residual small oscillations associated with the DB-only sensor drift.

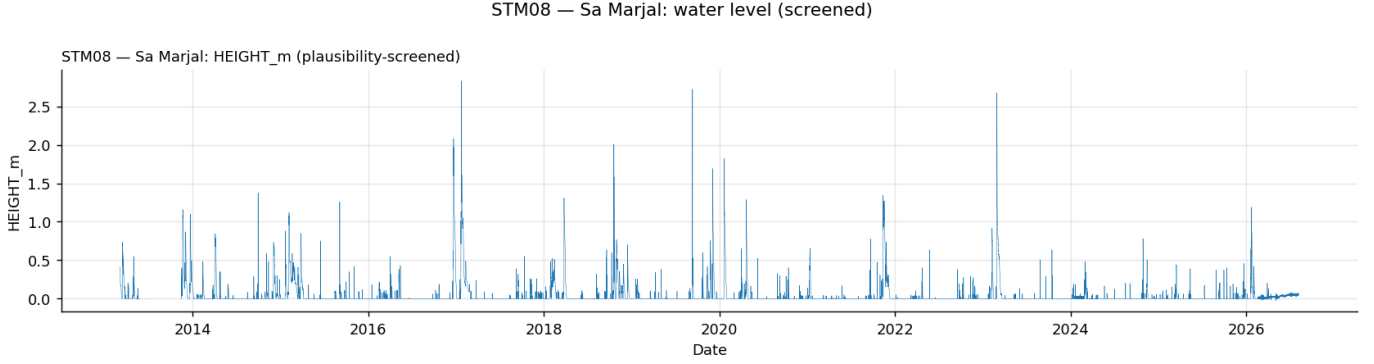


Figure 2. Screened STM08 water-level record used by the current pipeline. The series is highly intermittent, with isolated flood peaks separated by long near-zero periods.

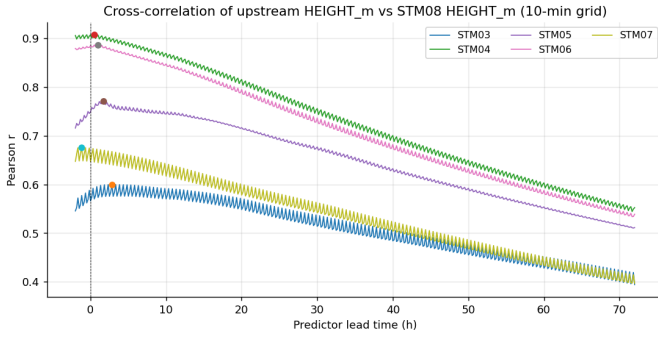


Figure 3. Historical cross-correlation between upstream hydrological levels and STM08. The strongest relationships occur at short routing lags.

The final catalogue contains 175 valid events: 102 in training, 16 in validation, and 57 in test. The mean duration is 34.9 hours, the median is 20.0 hours, and the maximum is 108 hours. The largest event reaches $87.84 \text{ m}^3 \text{ s}^{-1}$ and 2.83 m at STM08, during January 2017. Seventeen valid events have peak timestamps after the July 2025 extension boundary. Each event is assigned to one split using its peak timestamp; the event ID is then carried into the training table for per-event evaluation.

VI. DATA-DRIVEN BASELINES

A. Models

Six baseline models are evaluated for each of the three horizons.

- **Persistence:** the current STM08 level is used as the future level, $\hat{H}(t+h) = H(t)$.
- **Ridge:** regularised linear regression using the engineered predictors. Missing predictors are median-imputed using statistics fitted on the training split, and missingness indicators are retained.
- **Random forest:** 100-tree regression ensemble, using the training subsample that preserves event rows. The current implementation passes missing values natively.
- **XGBoost:** histogram-based gradient boosting with 300 trees, learning rate 0.1, maximum depth 8, and native missing-value handling.

- **ARIMA:** univariate STM08 model with order $(2, 1, 1)$ and three Fourier sine/cosine pairs representing the 24-hour cycle.
- **ARIMAX:** the same ARIMA specification with seven curated hydrological and rainfall exogenous variables. Future exogenous values are not observed during forecasting: they are carried forward from the forecast origin, avoiding temporal leakage.

The ARIMA order is selected by AIC between $(2, 1, 1)$ and $(1, 1, 1)$ on the last two years of the training split; $(2, 1, 1)$ is selected. ARIMA and ARIMAX are fitted on the training split and evaluated using daily rolling origins. At each origin, a 60-day observed window is used to update the state and a 288-step forecast covers all target leads within the following two days. This provides row-aligned predictions without using future observations.

B. Metrics

The primary metric is Nash–Sutcliffe Efficiency (NSE); values near one are best, while values below zero indicate performance worse than the observed mean. Kling–Gupta Efficiency (KGE), RMSE, MAE, and PBIAS complement NSE. Peak magnitude and timing errors and exceedance metrics (POD, FAR, and CSI) are also calculated at 0.5 and 1.0 m. Results are reported for the complete test split, its pre-extension validated portion, and the DB-extension portion. Per-event metrics are computed over the 57 test events.

C. Results

Table IV reports NSE. The test split includes the validated historical period and the corrected extension. The extension no longer produces the negative NSE collapse observed before the centimetre conversion.

Persistence is the strongest model by aggregate NSE at all three horizons. This is expected in an intermittent system: the level changes slowly during the dominant dry-state portion of the record. Ridge is the strongest non-persistence model by aggregate NSE, but its positive PBIAS increases with lead time. At 24 hours, full-test PBIAS is 70.3% for ridge, 116.2% for the random forest, and 138.7% for XGBoost. The tree models fit the training data very closely but generalise poorly

Table IV
TEST NSE BY MODEL AND HORIZON. EACH CELL CONTAINS FULL TEST / VALIDATED PORTION / EXTENSION PORTION.

Model	t+1 h	t+6 h	t+24 h
Persistence	0.988 / 0.990 / 0.979	0.913 / 0.915 / 0.900	0.563 / 0.536 / 0.707
Ridge	0.987 / 0.989 / 0.977	0.876 / 0.871 / 0.900	0.472 / 0.427 / 0.712
Random forest	0.982 / 0.984 / 0.973	0.761 / 0.753 / 0.802	0.292 / 0.301 / 0.241
XGBoost	0.972 / 0.973 / 0.964	0.721 / 0.694 / 0.866	0.091 / 0.017 / 0.486
ARIMA	0.856 / 0.852 / 0.875	0.777 / 0.764 / 0.845	0.455 / 0.417 / 0.653
ARIMAX	0.855 / 0.852 / 0.874	0.776 / 0.763 / 0.846	0.454 / 0.417 / 0.654

Table V
MEDIAN PER-EVENT NSE ON THE 57 TEST EVENTS.

Model	1 h	6 h	24 h
Persistence	0.184	-1.183	-2.305
Ridge	0.363	-1.962	-2.987
Random forest	0.297	-5.482	-14.533
XGBoost	0.314	-5.927	-10.876
ARIMA	-1.221	-0.731	-2.606
ARIMAX	-1.221	-0.733	-2.511

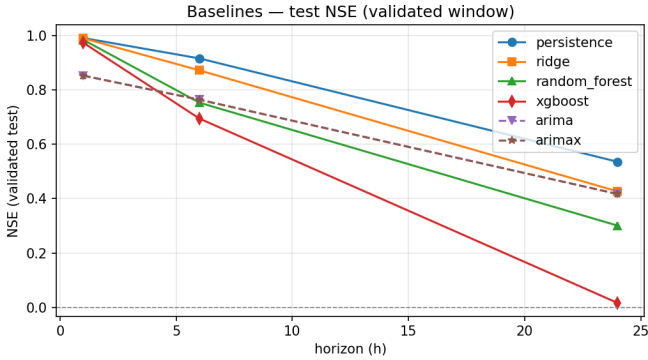


Figure 4. Test NSE on the validated portion of the test split. Performance decreases with forecast horizon for all baseline families.

at long horizons, indicating overfitting to the intermittent regime and the observed predictor distribution.

ARIMA and leakage-free ARIMAX have nearly identical results. The exogenous variables do not add measurable aggregate skill when their future values are not supplied unrealistically. ARIMAX has lower long-horizon bias than the supervised tree models, but it does not exceed persistence or ridge in NSE.

Per-event NSE is substantially lower than aggregate NSE, especially at 6 and 24 hours. This confirms that the rare flood peaks remain the difficult part of the task even after data-quality corrections. The event table is therefore more informative for model selection than aggregate scores alone, but its 57 test events still limit statistical certainty.

VII. PHYSICAL MODEL STATUS AND REMAINING SCOPE

The repository contains an initial HEC-HMS project, terrain and basin files, meteorological configurations, and a Java-based DSS interface. The basin delineation, parameter calibration, independent event validation, and uncertainty analysis have not yet been completed. Consequently, there is no HEC-HMS performance result in the current comparison.

The sequential data-driven phase is also pending. The next modelling stage is to train multi-horizon LSTM/GRU and Temporal Fusion Transformer models using the 10-minute training table, the missingness indicators, and the same chronological splits. Hyperparameter selection must use validation data only. Subsequent interpretability and comparison phases should include SHAP or an equivalent feature-attribution method, common event subsets, and a direct HEC-HMS versus machine-learning comparison.

The current baseline results have four important limitations. First, the target is highly intermittent, so aggregate NSE is dominated by ordinary low-flow periods. Second, long predictor gaps remain in the training table; tree models use native missing values while linear and ARIMA-family models use train-fitted imputation or carried-forward exogenous values. Third, quality-1 extension records are retained because their transitions are continuous, but they remain flagged as suspicious. Fourth, the diurnal drift correction is a data-quality rule based on historical comparison; it should be rechecked if new DB data or independent sensor information becomes available.

VIII. CURRENT CONCLUSIONS

The current project has reached a reproducible baseline stage. The most important data finding was that production-database water levels were in centimetres, while the modelling schema expected metres. Correcting this conversion restored realistic extension levels and rating-curve discharges. Quality-2 records were removed, quality-1 records were audited and retained, and an anomalous DB-only diurnal oscillation was corrected using a station-comparison rule applied uniformly across the hydro network.

The resulting catalogue contains 175 valid events after applying a global 0.10 m minimum peak-level criterion. Persistence remains the strongest aggregate baseline, which is an important result rather than a failure: it quantifies how difficult it is to outperform a state-continuity forecast in a mostly dry intermittent system. Ridge is the strongest non-persistence model by aggregate NSE, while ARIMA-family forecasts show comparatively controlled bias. None of the classical baselines provides consistently strong event-level performance at 24 hours.

These results establish the reference point for the remaining thesis work. The next scientific comparison must determine whether HEC-HMS process representation or sequential neural models can improve flood-event prediction without sacrificing the robust ordinary-condition behaviour of the persistence baseline.

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